

SUJITH KUMAR REDDY

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PROFESSIONAL SUMMARY

Backend-focused Software Engineer with 6.5+ years building Python and Java APIs, GraphQL services, and Postgres-backed systems at production scale. At Sona AI, designed GraphQL and REST APIs end-to-end, owning schema design and domain modeling for a platform serving 10K+ daily requests. At Rime Soft, built fault-tolerant, exactly-once Kafka event processing for a \$2B+ daily-volume financial platform. At HP, optimized PostgreSQL performance and deployed Kubernetes-based microservices at enterprise scale. Deep experience in domain modeling, distributed async processing, and writing robust, production-grade backend systems.

TECHNICAL SKILLS

- **Languages:** Python, Java, TypeScript, Go, Kotlin, C++
- **Backend and APIs:** FastAPI, Spring Boot, GraphQL, REST, gRPC, Microservices; domain modeling and OpenAPI contract-first design
- **Databases:** PostgreSQL, MySQL, DynamoDB, MongoDB, Redis, Aurora, Oracle; index redesign, partitioning, query optimization
- **Distributed and Async Processing:** Kafka, Spark Streaming, AWS Kinesis, Flink; event-driven pipelines and fault-tolerant, exactly-once delivery
- **Cloud and Infrastructure:** AWS (Lambda, S3, ECS, EKS, DynamoDB, API Gateway, CloudWatch), Kubernetes, Docker, Terraform, GitHub Actions
- **Frontend:** React (17+), Next.js, TypeScript; full-stack delivery across web and mobile clients
- **Observability and Quality:** Prometheus, Grafana, CloudWatch, ELK; JUnit, PyTest, TDD, 80%+ coverage enforcement

PROFESSIONAL EXPERIENCE

Sona AI

Austin, TX

Lead Software Engineer

June 2025 - Present

- Designed TypeScript GraphQL and REST APIs end-to-end, owning schema design, data models, and API contracts serving 10K+ daily requests at sub-100ms p95 latency.
- Sustained 99.9% uptime through 3x traffic growth by architecting Kafka event pipelines that decoupled ingestion from downstream services, enabling zero-downtime deployments.
- Cut backend latency by 70% by eliminating inefficient database queries, optimizing API execution paths, improving caching, and reducing unnecessary processing across the API layer.
- Automated CI/CD by codifying infrastructure with Terraform and parallelizing test stages in GitHub Actions, cutting deployment cycle time by 40% and enabling same-day production releases.
- Built Grafana and CloudWatch dashboards that tracked latency and error rates in real time, reducing incident MTTR by 30% through proactive alerting and anomaly detection.
- Designed domain-driven microservices architecture isolating orchestration, API gateway, and notification domains into independently deployable services, reducing cross-team coupling.
- Led cross-functional engineering design sessions and code reviews across teams in the United States and Vietnam, establishing shared coding standards that reduced PR cycle time and defect rates.
- Mentored junior engineers by conducting paired system design reviews and establishing PR review templates, accelerating onboarding and raising team-wide code quality.
- Built Sona AI's SaaS platform automating persona-based customer interactions end-to-end, cutting manual processing time by 35% through pipeline automation.
- Built a production LLM evaluation harness benchmarking OpenAI, Anthropic, and open-source models on latency, cost, and task accuracy, cutting LLM costs by roughly 30% versus a single-provider default.

HP

Bangalore, India

Senior Software Engineer

Aug 2021 - Dec 2022

- Engineered Java/Spring Boot microservices for HP's global firmware automation platform managing device lifecycle workflows across millions of enterprise printers and peripherals worldwide.
- Optimized PostgreSQL performance by redesigning indexes, partitioning large tables, and rewriting slow queries, accelerating real-time analytics and large-volume reporting across global infrastructure.
- Improved system reliability by 30% by chaining Kafka topics through AWS SQS and Redis caches into high-throughput event-driven pipelines.
- Instrumented services with Prometheus metrics and Grafana dashboards to surface request-level bottlenecks, reducing MTTR by 20%.
- Deployed services on AWS EKS using Kubernetes, Helm, and Argo CD for zero-downtime rollouts.
- Standardized API contracts across teams by adopting OpenAPI specifications and auto-generating client SDKs, reducing integration bugs by 20% across dependent service teams.
- Reduced the blast radius of firmware failures across HP's global device fleet by decomposing the legacy monolith into independently deployable Spring Boot microservices, cutting deployment lead time by 35%.
- Designed secure backend services with strict AWS IAM role boundaries, secret rotation via AWS Secrets Manager, and encrypted inter-service communication, meeting HP's enterprise security and compliance requirements.
- Collaborated with cross-functional product and firmware engineering teams across the US, India, and Europe in architectural design reviews, translating requirements into scalable backend service contracts.
- Automated test pipelines by writing JUnit integration tests and Testcontainers-based database fixtures, achieving 80%+ unit test coverage and catching data-layer regressions before production deploys.

Rime Soft Pvt Ltd

Hyderabad, India

Software Engineer

June 2019 - Aug 2021

- Built Java/Spring Boot and Python microservices powering Moody's financial payment and reconciliation platform, processing \$2B+ in daily transaction volume with sub-second settlement guarantees.
- Designed real-time data pipelines by routing Kafka events through Spark Streaming and AWS Kinesis, enabling consistent financial data delivery from ingestion to reporting with guaranteed ordering.
- Implemented Kotlin-based event consumers for real-time account reconciliation, processing high-frequency financial events with Kafka transactions to guarantee exactly-once delivery semantics.
- Boosted transaction throughput by 25% through composite index redesign, query rewrites, and Redis caching across PostgreSQL and DynamoDB; automated zero-downtime deploys with GitLab CI and Kubernetes.

- Migrated legacy payment monoliths into Spring Boot microservices by applying strangler-fig decomposition and defining OpenAPI contracts per domain, reducing deployment lead time by 40% and isolating failure boundaries.
- Productionized PyTorch-based fraud detection APIs by building a REST prediction-serving layer with A/B testing hooks, improving anomaly detection accuracy by 18% across payment flows.
- Collaborated with compliance and security teams to enforce PCI-DSS controls across payment APIs, implementing end-to-end encryption, input validation, and audit logging for all financial data flows.

Grepthor Technologies
Software Engineer

Hyderabad, India
Jan 2018 - June 2019

- Built Python/FastAPI services with C++ performance modules for HDFC Bank's digital platform, delivering secure, low-latency banking workflows to 1M+ customers with OAuth2/JWT multi-tenant authentication.
- Designed real-time Kafka/Flink ingestion pipelines that ensured consistent event ordering and data quality across banking transaction flows; automated releases with Jenkins, Kubernetes, and Terraform.
- Improved MySQL performance by analyzing slow query logs, adding composite indexes, rewriting N+1 queries, and introducing connection pooling, increasing transaction throughput by 20%.
- Implemented OAuth2 authorization flows and JWT token validation by building a centralized auth service shared across multi-tenant banking integrations, securing both internal and external API access.
- Automated deployments by writing Jenkins declarative pipelines that built Docker images, ran integration tests, and applied Kubernetes manifests with Terraform-managed infrastructure, reducing release effort by 30%.
- Accelerated compute-heavy banking calculations by writing C++ extension modules called from Python FastAPI handlers, reducing API response time for high-volume transaction queries by 40%.
- Coordinated with HDFC security and compliance teams to implement API rate limiting, request validation middleware, and audit logging across all customer-facing banking endpoints.

PROJECTS

- StatRush - AI-Powered NBA Analytics Platform (Jan 2023 - Present) |** statrush.vercel.app | github.com/sujithreddyp0123
- Built full-stack SaaS analytics platform with React/Vite frontend, Python/FastAPI backend, PostgreSQL, and Redis deployed on Vercel and Render.
 - Engineered XGBoost/LightGBM ML ensemble with live DraftKings prop line integration via The Odds API - real-time inference at sub-200ms latency with explainable confidence scoring.
 - Implemented Kelly Criterion bet sizing and edge calculation to surface high-value prop opportunities with feature attribution dashboards.

- Agri AI - Telugu-First Farming Assistant for Andhra Pradesh and Telangana**
- Built an India-focused, Telugu-first farming assistant helping farmers with crop photo diagnosis, fertilizer advice, weather-aware recommendations, and season tracking.
 - Built with a React/Vite frontend, FastAPI backend, and SQLite database deployed on Render, integrating Claude Vision for crop image analysis.
 - Designed an ML/XGBoost-ready data layer to support future crop-risk and recommendation models.

EDUCATION

- **Master's in Information Systems** - University of Houston-Clear Lake (Jan 2023 - Dec 2024)
- **Bachelor's in Computer Science** - Panimalar Institute of Technology, Anna University (2014 - 2018)

CERTIFICATIONS

- AWS Certified Developer - Associate (Issued Apr 2025, Expires Apr 2028)
- Certified ScrumMaster (CSM)